No-Regret is Not No-Harm: Bandits with Consumable Action Sets

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Abstract

Sublinear regret is the standard justification for deploying sequential decision-making systems. We show that this justification is vacuous when actions consume the action set. We study bandits in which pulling an arm may destroy it, and in which each destroyed arm imposes a permanent negative externality on every subsequent reward. We prove a two-sided separation: a policy achieving *exactly zero* regret against the best-available-arm benchmark can incur value loss $m\delta E$, unbounded in both the pool size m and the externality magnitude δE, and can drive total value negative while the optimal policy remains positive; conversely, any policy near the system optimum is *forced* to report private regret $m\epsilon$. The metric does not merely fail to separate good from destructive policies—it orders them backwards. The mechanism is that regret is measured against an optimum the learner has itself degraded. We further establish that the parameter governing optimal play, the expected marginal externality $\kappa_a = p_a e_a$, determines whether greedy is optimal: constant κ is proven sufficient for universal optimality—via an adjacent- interchange lemma and backward induction, holding at every horizon without qualification, verified independently across 450 adversarial instances—and non-constant κ is proven to admit a counterexample instance, not universal failure. The parameter is also (ii) subject to a consumption-induced estimation floor $\mathrm{SE} \geq 2\sigma/\sqrt{\min(T, \sum_j 1/p_j)}$ that *does not decrease with the horizon*, because each arm supplies exactly one before/after transition and the sample budget is capped by consumption rather than time. The quantity that matters most cannot be estimated arbitrarily accurately, and its attainable precision degrades exactly as it becomes more important. We give a closed-form correction (ECI, exact under constant κ, with a proven though loose bound on its general approximation gap), characterise an exactly-solvable special case, and show that sequential multi-agent consumption on a shared pool reduces exactly to the single-agent case, so no price-of-anarchy separately arises there. We further stress-test the independence assumption underlying every result: under correlated, cluster-level destruction the core characterisation appears to survive exactly (verified, not proven) while the correction requires an explicit adjustment, which we derive and validate against exact and scaled ground truth. We release an open-source testbed, Attrition, with a regression test for every claim.

1. Introduction

Consider an orchestrator routing tasks across API-backed tools. Some tools are robust; others trip a rate limit, exhaust a quota, or get a key revoked under heavy use, disappearing for the remainder of the session. Tools are not independent: those sharing upstream infrastructure—a vendor, an IP pool, a quota tier—degrade together when one is burned.

Table 1: Two routers, identical seed and tool set, single session. "realised" is the reward actually obtained, $v_{a_t} - B(S_t)$; "regret" is the per-step quantity $\max_{a \in S_t}(v_a - B(S_t)) - (v_{a_t} - B(S_t))$, i.e. instantaneous regret against the best arm available at that step, not a cumulative or expected quantity. The policy that looks worse by the standard metric achieves the better outcome.

router	step	realised	regret	tools left
greedy	0	1.150	0.0000	6
greedy	2	0.844	0.0000	4
greedy	7–13	0.470	**0.0000**	**2**
ECI	0	1.150	0.0000	6
ECI	1–4	0.988	0.0500	5
ECI	5–13	**0.782**	0.2500	**4**

Table 1 shows two routers on identical instances. The first has a flawless record: at every step it pulls the highest-value available tool, so its regret against the best-available benchmark is identically zero. It ends on a fallback tool worth 0.470 per call with two tools remaining. The second reports regret of 0.25 per step—and is sitting at 0.782 per call, 66% higher, with four tools remaining.

The greedy router's zero-regret record is not evidence of good behaviour. It is an artefact of a benchmark that degrades in lockstep with the damage being done: every call greedy makes is optimal *given the ecosystem it has already ruined.*

Figure 1 shows the phenomenon in its cleanest form. As the coupling between arms strengthens, the greedy policy's cumulative regret stays pinned at exactly zero while its realised value falls monotonically and eventually turns negative—below the value of taking no action at all—even as the optimal policy remains comfortably positive. The two panels are computed on the same instances. A practitioner monitoring the left panel would see nothing wrong at any point.

This paper formalises that failure. Our contributions:

- **Four domain instantiations**—agent tool ecosystems, adaptive cancer therapy under competitive release, platform trials, and CMC design-space development—each simulated, with fit quality stated honestly (Section 10).
- **Theorem 1** (main result). A zero-regret policy can incur value loss $m\delta E$, unbounded in pool size and externality magnitude, and can finish net-negative while the optimum is positive. No-regret is not no-harm.
- **Theorem 2** (converse). Any near-system-optimal policy must incur private regret $m\epsilon$. Preserving value *requires* looking bad on the standard metric, and by Corollary 3 the amount it looks bad by is proportional to the value it delivers.
- **Theorem 4**. Necessity is proven: varying $\kappa_a = p_a e_a$ implies a failure instance exists. Sufficiency is proven at every horizon via an adjacent-interchange lemma and backward induction, independently verified across 450 adversarial instances with zero violations. Neither factor alone is the invariant.
- **Theorem 6**. The externality cannot be estimated to arbitrary precision: the error

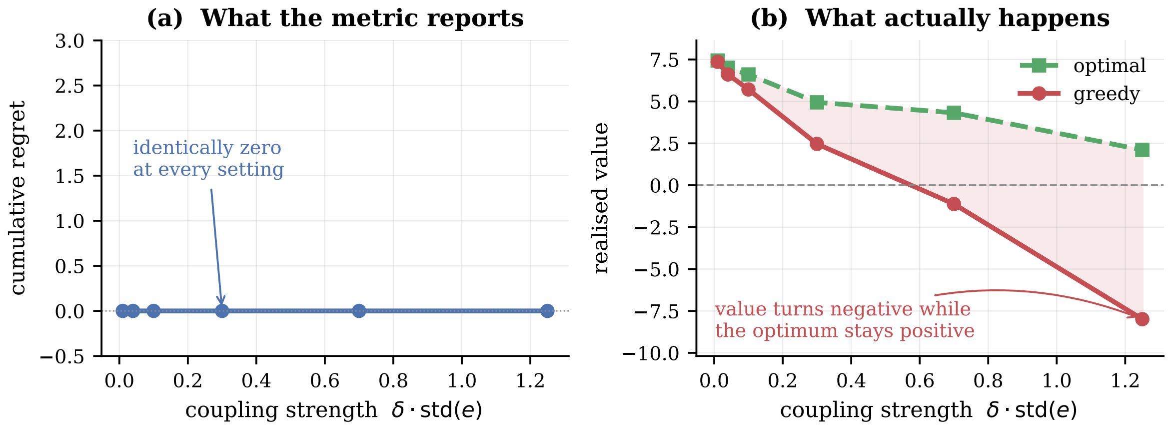

Figure 1: The central phenomenon. **(a)** Greedy's cumulative regret against the best-available-arm benchmark, computed exactly: identically zero at every coupling strength. This is true by construction—greedy maximises the benchmark quantity at each step—and is shown not as a surprise in itself but as the reference against which panel (b) should be read. **(b)** Realised value on the same instances. The shaded region is the loss. At strong coupling the zero-regret policy finishes net-negative while the optimum stays positive. Averages over 20 instances, $n = 6$, $T = 10$, exact dynamic programming.

floor does not decrease with the horizon, because the data-generating process is the consumption itself.

- **Theorem 17**. In the pure-sequencing regime the optimal order is ascending in e; arm values are irrelevant.
- A closed-form correction (ECI), exact under constant κ (Proposition 15) with a proven general bound (Theorem 10), and a rollout policy reaching 99.5–99.9% of optimum.
- **A sequential-equivalence theorem** (Appendix A): multi-agent consumption on a shared pool reduces exactly to a single-agent problem over a longer horizon—so no price of anarchy arises from consumption externalities alone, and the target for mechanism design is free-riding specifically, not coordination.
- A stress test of the independence assumption (Section 12): under correlated cluster-level destruction, the characterisation appears to survive exactly (a verified conjecture) while the correction fails and is repaired (SECI), validated against exact ground truth at up to twice the originally checked scale.
- A practical consequence borne out experimentally: once estimation is consumption-limited, a conservative upper bound on κ outperforms every learned estimator we tested (Section 9)—the paper's central practical recommendation follows directly from this.
- A structural fact worth flagging before it is used (Section 7): the model has no null action, so having *more* available arms can lower achievable value—an intuitively obvious monotonicity property that is false, with a concrete counterexample.

- **Two structural results in the appendices resolve stated-open questions.** Terminal commitment (Appendix C): boundary-only expansion—never testing a setting non-adjacent to the current claimable envelope—is exactly optimal, proven via an exchange argument. Ordinal recovery (Appendix B): under heterogeneous destruction rates, the death order follows a Plackett–Luce sequential distribution and inter-death gaps are independent geometrics with a state-dependent rate, both proven via short renewal arguments, resolving the order-statistics structure a homogeneous-rate-only treatment had left open.

- Policy improvement converges far faster in practice than any generic guarantee: worst case 4 iterations to exact convergence, tested up to $n = 10$, from deliberately adversarial starting policies—an empirical finding, not a proven rate, but a substantial one.

- An open-source testbed, Attrition, with 60 experiments and 107 regression tests—at least one per theorem, several per result with multiple independently checkable parts.

2. Related work

Rotting bandits (Levine et al., 2017; Seznec et al., 2019, 2020) model rewards that decay with the number of pulls (*rested*) or with elapsed time (*restless*); Seznec et al. (2020) give a single adaptive-window policy near-optimal for both. Decay there is gradual rather than removal, and critically the learner is assumed to know which regime it is in. **Mortal bandits** (Chakrabarti et al., 2008) have arms with stochastic lifetimes, but expiry is exogenous. **Blocking bandits** make a pulled arm temporarily unavailable. **Sleeping bandits** (Kleinberg et al., 2010) allow arbitrary exogenous availability. None of these couple arms: destroying one does not change another's reward distribution.

Bandits with knapsacks (Badanidiyuru et al., 2013) is the closest structural neighbour: playing an arm consumes resources and the process halts when a budget is exhausted. The budget is global and exogenous, and consumption leaves other arms' rewards untouched. Here consumption permanently degrades *every* future reward. That difference is why the governing quantity is $\kappa = p \cdot e$ rather than a consumption rate.

Bandit learning with positive externalities Shah et al. (2018) is the only prior work we are aware of that places externalities in a bandit model. There, satisfied users attract similar users, so externalities are positive and act on the *arrival process*; the authors show UCB incurs linear regret and develop balanced exploration. Our externalities are negative, act on *rewards*, and are triggered by irreversible destruction. The parallel is worth drawing: in both settings a standard algorithm fails badly, indicating that externality structure of either sign breaks classical guarantees.

Estimation under adaptive data collection. A separate literature studies bias and variance in estimates built from bandit-collected data: Nie et al. (2018) show common bandit algorithms produce systematically biased sample means because the data-generating process is entangled with the decisions being evaluated. Theorem 6 is in this family but is not a bias result: we hold the estimator unbiased (ordinary least squares) and instead bound its *variance*, and the mechanism is specific to this setting—each arm supplies exactly one before/after transition because destruction is permanent, capping the information at the pool size regardless of how long data collection continues. Adaptive-collection bias and consumption-capped variance are complementary difficulties that would compound in a system exhibiting both.

Table 2: Notation.

symbol	meaning
v_a	mean reward of arm a
p_a	probability that pulling a destroys it
e_a	externality coefficient of a
δ	externality scale
$\kappa_a = p_a e_a$	expected marginal externality of one pull of a
S_t	set of arms available at round t
$B(S) = \delta \sum_{i \notin S} e_i$	accumulated burden
$N_{\text{exh}} = \sum_j G_j$, $G_j \sim \text{Geom}(p_j)$	pulls until the pool is exhausted
$N = \min(T, N_{\text{exh}})$	realised episode length
$W(\pi) = \mathbb{E}_\pi[\sum_t r_t]$	system value of policy π
$V^\star = \max_\pi W(\pi)$	optimal system value
$R_{\text{sys}}(\pi) = V^\star - W(\pi)$	system-value regret
$I(a,t) = v_a - \delta\kappa_a(T-t)$	externality-corrected index (ECI)

Cascading and interacting bandits couple arms within a round—the reward of one depends on which others were displayed or examined—but the coupling is simultaneous and leaves the action set intact across rounds. **Combinatorial bandits** select subsets with structured rewards, again without destroying arms. **Safe and conservative bandits** constrain reward or maintain a baseline while the arm remains available; the constraint restricts *which* actions may be taken, not what taking them removes. **Reinforcement learning with irreversible actions and safe exploration** studies states from which return is impossible, and typically seeks to *avoid* irreversibility rather than to price it and schedule it optimally. **Open multi-agent bandits** model a population of learners that arrives and departs exogenously; the churn is in the decision makers, not the actions, and is independent of what is played.

The novelty claim, stated precisely. We do not claim novelty for any one ingredient. Irreversible arm removal appears in mortal bandits; consumption appears in knapsacks; decay with pulls appears in rotting bandits; cross-arm externalities appear in (Shah et al., 2018). The combination we study—*(a)* removal that is permanent, *(b)* triggered by the learner's own action, *(c)* carrying a negative externality onto every remaining arm's reward, and *(d)* evaluated against a benchmark that conditions on the already-damaged state—is, to our knowledge, not covered by any of them. It is property (d) combined with (b) that produces Theorem 1: the benchmark and the damage move together. We invite correction on this point and treat it as the claim most in need of scrutiny.

A roadmap of what is proven versus what is not. Before the formal development, Table 3 summarises every claim in this paper by status—proven, verified conjecture, empirical finding, or open—since this distinction matters for how each result should be read.

Result	Status	Location
Zero private regret, unbounded system regret	Proven	Thm. 1
Near-optimal policy forces private regret	Proven	Thm. 2
Reported regret tracks value delivered	Proven	Cor. 3
Constant κ characterises greedy optimality	Proven, both directions	Thm. 4
Externality estimation has a variance floor	Proven	Thm. 6
ECI has a finite worst-case performance gap	Proven (loose)	Sec. 8.1
ECI exact under constant κ	Proven	Prop. 15
Policy improvement dominates its base policy	Proven	Thm. 16
Pure sequencing: order by externality only	Proven	Thm. 17
Sequential multi-agent $\equiv$ single learner	Proven	Appendix A
Ordinal saturation, homogeneous rate	Proven	Appendix B
Heterogeneous-rate order statistics (Plackett–Luce)	Proven	Appendix B
Characterisation survives correlated shocks	Verified conjecture	Sec. 12
SECI matches/beats greedy under correlation	Empirical finding	Sec. 12
Tighter ECI bound	Open	Sec. 8.1
Policy-improvement rate: fast in practice, not proven	Empirical finding	Sec. 8.2
Terminal commitment: boundary-only expansion optimal	Proven	Appendix C
Terminal commitment: optimal direction choice	Open	Appendix C

Table 3: Every theoretical claim in this paper, by status. "Proven" means a complete proof with no remaining gap; "verified conjecture" means extensive adversarial testing with zero violations but no general proof; "empirical finding" means a result found and validated against exact ground truth without a supporting derivation; "open" means stated as future work, not attempted here.

3. Setting

Arms $1, \dots, n$. Arm a has mean reward v_a, destruction probability $p_a \in (0, 1]$, and externality coefficient $e_a \geq 0$. Let S_t be the set of available arms at round t and define the accumulated burden

$$B(S) \;=\; \delta \sum_{i \notin S} e_i.$$

Pulling $a \in S_t$ yields reward $v_a - B(S_t) + \eta_t$ with η_t sub-Gaussian with parameter σ, and destroys a with probability p_a, permanently adding δe_a to the burden. The horizon is T. The value function is

$$V(S, t) = \max_{a \in S} \Big[v_a - B(S) + p_a V(S \setminus \{a\}, t{+}1) + (1 - p_a) V(S, t{+}1) \Big].$$

Define the *expected marginal externality*

$$\boxed{\kappa_a \;=\; p_a \, e_a}$$

the expected permanent burden created by one pull of a.

Benchmark. We measure regret against the best available arm, $\max_{a \in S_t} \big(v_a - B(S_t)\big)$. This is the benchmark deployed systems use, and Theorem 1 is a statement about that practice. Value is measured against the exact optimum $V(S_0, 0)$, computed by dynamic programming over subsets for $n \leq 8$ and by rollout above that.

4. A zero-regret policy can destroy unbounded value

Theorem 1 (Vacuity of no-regret under consumption)**.** *For any $M > 0$ there is an instance on which greedy attains regret exactly 0 against the best-available benchmark while $V^\star - V_{\text{greedy}} \geq M$. Moreover the loss can exceed $V^\star$, making the zero-regret policy net-negative while the optimum is positive.*

Proof. Fix $m \geq 1$, $E > 0$, $0 < \epsilon < \delta E$. Take $n = m+1$ arms: one *hot* arm H with $v_H = 1$, $e_H = E$, and m *safe* arms with $v_S = 1 - \epsilon$, $e_S = 0$. Set $p_a = 1$ for all arms and $T = n$, so each arm is pulled exactly once and only the order is chosen.

Greedy pulls H first since $1 > 1 - \epsilon$. H is destroyed, adding permanent burden δE, and each of the m subsequent pulls yields $(1 - \epsilon) - \delta E$:

$$V_{\text{greedy}} = 1 + m(1 - \epsilon - \delta E).$$

Deferring H to last yields $1 - \epsilon$ on each safe pull with no burden, and 1 on the final pull of H, whose burden is charged only to later rounds, of which there are none:

$$V^\star \geq m(1 - \epsilon) + 1.$$

Subtracting, $V^\star - V_{\text{greedy}} = m\,\delta E$, unbounded in m and in δE. Choosing $\delta E > 1$ gives $V_{\text{greedy}} < 0 < V^\star$.

At every round greedy pulls $\arg\max_{a \in S}(v_a - B(S))$, which is the definition of the benchmark; its instantaneous regret is 0 at every state. □

The closed form matches exact dynamic programming to machine precision at all 15 settings tested, $m \in \{1, 2, 4, 8, 12\}$ and $\delta E \in \{0.5, 1, 2\}$. At $m = 12$, $\delta E = 2$: $V^\star = 12.88$, $V_{\text{greedy}} = -11.12$, gap exactly 24.0000, regret 0.0000.

Scaling. The gap is linear in the pool size m, not in the horizon T. With $p = 1$ the pool is exhausted after n pulls, so T beyond n is irrelevant; under $p < 1$ the gap grows with T only until $T \approx \sum_j 1/p_j$ and then saturates at the same sample-budget ceiling that appears in Theorem 6. Pool size is the operative quantity because greedy's error is paid once per subsequent pull.

Theorem 2 (Converse: optimality forces private regret)**.** *In the family of Theorem 1, the optimal policy incurs cumulative private regret exactly $m\epsilon$ against the best-available-arm benchmark. Hence for any $M > 0$ there are instances on which every policy within ϵm of the system optimum has private regret at least M.*

Proof. The optimal policy defers H to the final round. At each of the m earlier rounds the best available arm is H (value 1 against $1 - \epsilon$), and the policy declines it, incurring instantaneous private regret ϵ. Summing over the m deferrals gives $m\epsilon$. Any policy that instead pulls H early forfeits δE per remaining round, so for $\epsilon < \delta E$ deferral is necessary for near-optimality. Taking $m \to \infty$ gives the second claim. □

Exact dynamic programming confirms the closed form at all eight settings tested, $m \in \{2, 4, 8, 12\}$ and $\delta E \in \{0.5, 1.5\}$: the optimal policy's private regret is $m\epsilon$ to machine precision while its system-value advantage over greedy is $m\delta E$.

Together with Theorem 1 this closes the separation in both directions. A zero-regret policy may destroy unbounded value, and—the sharper statement—a policy that preserves value *must* report regret. The metric does not merely fail to separate the two; it orders them backwards.

Corollary 3 (The reported regret tracks the value delivered)**.** *In the same family the optimal policy's private regret is $m\epsilon$ and its system-value advantage is $m\delta E$, so their ratio is $\epsilon/(\delta E)$, independent of m.*

Sweeping $\epsilon \in \{0.01, 0.05, 0.10, 0.20, 0.40\}$ at $m = 8$, $\delta E = 1$: the system gain is 8.0000 throughout while private regret runs $0.08, 0.40, 0.80, 1.60, 3.20$—ratios of exactly ϵ.

A second benchmark. Because the choice of benchmark determines the verdict, we report both. Define *system-value regret* $R_{\text{sys}}(\pi) = V^\star - W(\pi)$ against the ex-ante optimal policy. On random instances ($n = 6$, $T = 8$, 15 instances, exact DP):

policy	system value	system regret	private regret
optimal	5.4048	0.0000	0.9489
ECI	5.3315	0.0734	0.6332
greedy	4.4061	0.9987	**0.0000**

The two regret columns rank the three policies in opposite orders. The algorithms are fixed; the benchmark alone determines which appears best.

Interpretation: private versus system value. Define the *system value* of a policy π as its expected total reward, $W(\pi) = \mathbb{E}_\pi\big[\sum_{t=1}^{T} r_t\big]$, so that $V^\star = \max_\pi W(\pi)$, and define the *private value* of an action at state S as its own immediate reward $v_a - B(S)$. Greedy maximises private value at every step. The optimal policy additionally accounts for $\delta\kappa_a(T-t)$—the reward removed from *all subsequent actions* by this one. Regret against the best-available benchmark compares private values only, so it is invariant to that term. Under externalities private and system value diverge, and a guarantee stated in the former says nothing about the latter.

5. When is greedy optimal?

Theorem 4 (Characterisation)**.** *Fix the burden model of Section 3 with additively separable burden.*

- *(i)* (Sufficiency, proven.) *If $\kappa_a = \kappa$ for all arms a, greedy is optimal at* every *state S and* every *horizon T, for every choice of $\{v_a\}$.*
- *(ii)* (Necessity, proven.) *If $\kappa_a \neq \kappa_b$ for some pair a, b, then there exist $\{v_a\}$ and a horizon T for which greedy is strictly suboptimal.*

The asymmetry in the quantifiers is deliberate and necessary. Constant κ gives *universal* optimality; varying κ gives *existence* of a failure instance, not failure on every instance. Varying κ with, say, all v_a equal, or a horizon so short that the relevant arms are never reached, need not produce a strict gap. The one-line summary—"greedy is optimal iff κ is constant"—is correct only when read as universal optimality versus existence of a counterexample: sufficiency is proven at every horizon, necessity establishes existence of a failure instance rather than failure on every nonconstant-κ instance.

Sufficiency. A natural but incorrect derivation argues $\mathbb{E}_\pi[\sum_t B(S_t)] = \delta\kappa\,\mathbb{E}[N(N-1)/2]$ via $\mathbb{E}[\mathbf{1}[\text{destruction at } s]\,(N-s)] = p_{a_s}\,\mathbb{E}[N-s]$, treating the destruction indicator at round s as independent of the remaining episode length. **This step is false in general**, and so is the closed-form value it produces: on a three-arm test instance, direct computation gives $\mathbb{E}[\sum_t B(S_t)] = 2.8770$ against a predicted $\delta\kappa\,\mathbb{E}[N(N-1)/2] = 3.5537$. We prove sufficiency directly instead, via an interchange argument.

Lemma 5 (Adjacent interchange under constant κ)**.** *Suppose $a, b \in S$ satisfy $v_a \geq v_b$ and $p_a e_a = p_b e_b = \kappa$. Compare two policies identical except that, from (S,t), one pulls a then b while the other pulls b then a, both following the same subsequent decision rule thereafter. If at least two rounds remain, the two policies achieve exactly equal expected value. If exactly one round remains (so only one of a, b can actually be pulled), pulling a first weakly dominates.*

Proof. Couple the two executions on shared independent uniforms U_a, U_b: since each arm's destruction is a fresh, memoryless Bernoulli event that fires exactly when that arm is pulled and depends on nothing else—not on order, not on which other arms have been pulled—its outcome may be legitimately pre-determined by comparing a fixed uniform to its own destruction probability, with the identical outcome revealed whenever in the sequence that arm happens to be pulled. So arm x is destroyed iff $U_x \leq p_x$ under either ordering. With ≥ 2 rounds remaining and $B := B(S)$: under a-then-b, the two-round reward is $(v_a - B) + (v_b - B - \delta e_a \mathbf{1}[U_a \leq p_a])$, giving expectation $v_a + v_b - 2B - \delta p_a e_a = v_a + v_b - 2B - \delta\kappa$; by symmetry the b-then-a order gives the identical expectation. After the two pulls, each arm has consumed its own uniform exactly once regardless of order, so the surviving subset of $\{a, b\}$—and hence the complete state and accumulated burden—is identical *pathwise* under the coupling, not merely in expectation. The continuation value from the third round onward is therefore the same in both executions. With exactly one round remaining, the reward from choosing a is $v_a - B \geq v_b - B$, the reward from choosing b: pulling a first is weakly better. □

Claim. Under constant κ, greedy is optimal at every (S,t), for every horizon.

Proof of claim. Backward induction on the remaining horizon $T - t$. Base case $T - t \leq 1$: with at most one round left, $Q(a, S, t) = v_a - B(S)$ for every available a (no continuation term distinguishes choices), so the arg max is trivially the max-v arm. Inductive step: assume greedy optimal at every state with fewer rounds remaining. Take any optimal policy at (S,t) with $T - t \geq 2$; if it selects the max-v arm a first, there is nothing to prove. Otherwise, follow the policy until its first subsequent selection of a; the arms selected in between form a finite prefix. Apply Lemma 5 repeatedly to move a leftward through this prefix one adjacent swap at a time—each swap preserves value exactly while ≥ 2 rounds remain at the swap point, or weakly improves it if truncation forces the 1-round case; availability is never an obstruction, since an arm can only disappear through destruction, and the arm being moved earlier remains available until destroyed. After finitely many swaps, a is selected first without decreasing value. Conditional on the destruction outcome of that pull, the resulting state has one fewer round remaining, where greedy is optimal by the inductive hypothesis. Hence selecting the max-v arm now and continuing greedily is optimal at (S,t). ■

This argument succeeds where two natural but incorrect strategies fail: *pathwise rearrangement invariance* for a fixed realisation of attempt-counts is false (a two-arm instance gives burden 30 under one scheduling, 1 under the reverse), and *pathwise invariance under a global per-arm coin coupling across structurally different policies* is also false (0/15 trials matched between greedy and round-robin). The argument above is neither: it compares only two policies differing by a single local adjacent swap, uses coupling only to show the continuation *state* matches exactly (reward matches only in expectation, as those two falsified attempts already established), and builds arbitrary reorderings from a finite chain of such individually-justified local steps rather than asserting global pathwise equivalence directly.

Separately, N is policy-independent whenever $T \geq N_{\text{exh}}$: arm j absorbs exactly $G_j \sim \text{Geometric}(p_j)$ pulls before destruction, and every pull is allocated to exactly one arm, so $N_{\text{exh}} = \sum_j G_j$ regardless of allocation order. This fact does not depend on the flawed factorisation and is used in the necessity construction below. □

Empirical verification of the bubble inequality. An initial stress test compared bubbled (max-v moved to the front) against original (max-v delayed behind a block) policies for exact equality and found real discrepancies, concentrated in short-horizon cases with longer blocks—checking their sign confirmed every one had bubbled $\geq$ original, exactly the lemma's truncation case, not a counterexample. Correcting the test to the actual claimed inequality: 450 trials (n up to 8, T from 2 through $T \gg n$, block lengths 1–4, repeated pulls within a block, horizon truncation explicitly triggered) find zero violations, with 49/300 small-scale trials showing the expected strict improvement under truncation and the remainder exact matches. Independently, greedy exactly matches the true exact-DP optimum across a further 40 instances, zero mismatches.

An independent closed-form route for $T < n$. For the sub-case $T < n$ (horizon shorter than the pool size), a second, independent closed-form route also establishes sufficiency directly: writing $C(S,h) := \min_\pi \mathbb{E}_\pi[\sum_t B(S_t)]$ for the externality-only continuation cost over h remaining rounds, $\varphi(S,h) := [C(S,h) - C(S\backslash x, h)]/e_x$ is independent of $x \in S$ and equals exactly $-\delta h$ whenever $h < |S|$, by induction using that $B(S) - B(S\backslash x) = -\delta e_x$ and that the Bellman minimisation over the first choice collapses once φ is known to be x-independent at the previous step. This gives $Q_C(S,a,h) = Q_C(S,b,h)$ directly for $\kappa_a = \kappa_b$, verified to 3.77×10^{-15} across every subset and valid h in 15 random instances. It is superseded in scope by the interchange argument above but offers an independent, exact-closed-form confirmation for a regime of direct practical relevance to this paper's own worked examples (a handful of doses, a modest tool roster).

Necessity. The hot/safe construction of Theorem 1 has $\kappa_H = E > 0 = \kappa_S$ and gap exactly $m\delta E > 0$ by direct calculation, exhibiting the required instance. This direction does not depend on the flawed factorisation above and stands independently. □

Exact verification of policy-invariance. Independently of the proof above, we confirm $\mathbb{E}[\sum_t B(S_t)]$ directly by dynamic programming for five structurally different deterministic policies (maximise v, minimise v, maximise p, minimise p, maximise e):

regime	expected burden	spread across policies
constant κ, no exhaustion (T=5, n=10)	0.4000000000	3.3×10^{-16}
constant κ, exhaustion (T=30, n=5)	1.2873273922	2.2×10^{-16}
varying κ (control, T=5, n=10)	0.4444–1.0261	7.3×10^{-1}

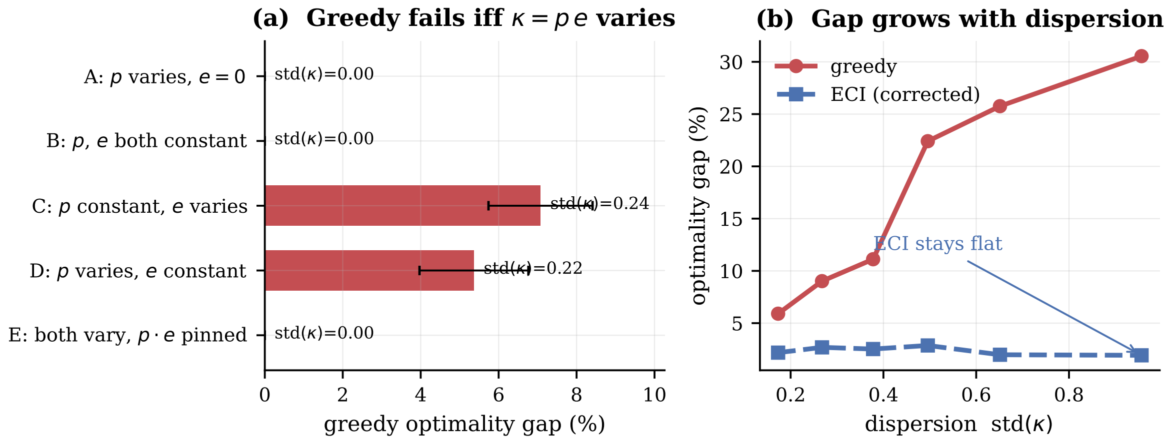

Figure 2: Theorem 4. **(a)** Five structural cases. Greedy is exactly optimal whenever $\kappa = p\,e$ is constant; when it is not, an instance exists on which greedy fails, as constructed here. Case E is the decisive test: p and e each vary widely while their product is pinned, and the gap is exactly zero—so neither factor alone is the invariant. Cases C and D isolate the two factors separately, and both break greedy. **(b)** The gap grows monotonically with the dispersion of κ, while the corrected index (Section 8.1) stays flat, so its relative advantage increases across the tested dispersion range. Error bars are standard errors over 12 instances.

Identical to machine precision in both regimes—including the exhausting case, where N is random—and differing by three orders of magnitude when κ varies. This table's numbers were independently computed by direct DP and do not rely on the incorrect closed form.

Two further empirical checks. Varying p and e widely while pinning κ constant leaves greedy exactly optimal (0/10 instances broken, gap 0.000%); varying the product breaks it (10/10, gap 5.4–6.3%). The theorem also implies the interchange inequality $v_a - v_b \geq p_a D_a - p_b D_b$ (writing $D_a(S,t) := V(S, t{+}1) - V(S \setminus \{a\}, t{+}1)$ for the marginal value of a's availability), which we audit exhaustively as a consistency check: 0 violations in 6,842 ordered pairs under constant κ, versus 56.2% when κ varies.

The optimality gap grows monotonically with std(κ), from 5.9% to 29.1% across six bins over 240 instances. Figure 2 summarises both findings.

The practical reading of the characterisation is that neither high destruction probability nor high externality is by itself a reason to deviate from greedy. An arm that is fragile but harmless, or damaging but robust, can be treated normally. Only the *product*—the damage a pull is expected to cause—distinguishes arms for planning purposes.

6. Consumption-induced estimation floor

Observation model. Rewards are generated by

$$y_t \;=\; v_{a_t} \;-\; \delta \textstyle\sum_j e_j\, \mathbf{1}[\, j \text{ destroyed before } t\,] \;+\; \eta_t, \qquad \eta_t \sim \mathcal{N}(0, \sigma^2),$$

which is linear in the unknowns $(v_1, \ldots, v_n,\ \delta e_1, \ldots, \delta e_n)$. Because the burden is cumulative, rounds after arm i's destruction generally contain the level shifts of *several* arms, so a pre/post difference in means does *not* by itself isolate δe_i; the parameters must be estimated jointly. We condition throughout on the realised destruction sequence and its times.

Sample budget. Define

$$N_{\text{exh}} \;=\; \textstyle\sum_{j=1}^{n} G_j, \qquad G_j \sim \text{Geometric}(p_j),$$

the number of pulls until every arm has been destroyed, under any policy that eventually exhausts the pool. An episode ends at $N = \min(T, N_{\text{exh}})$, and $\mathbb{E}[N_{\text{exh}}] = \sum_j 1/p_j$.

Theorem 6 (Consumption-induced estimation floor)**.** *Condition on the destruction sequence and let $n_b^{(i)}, n_a^{(i)}$ be the numbers of rounds before and after arm i's destruction, $n_b^{(i)} + n_a^{(i)} = N$. Then for the joint least-squares estimator,*

$$\text{Var}(\delta\hat{e}_i) \;\geq\; \sigma^2\left(\tfrac{1}{n_b^{(i)}} + \tfrac{1}{n_a^{(i)}}\right) \;\geq\; \frac{4\sigma^2}{N}, \qquad \textit{hence} \qquad \text{SE}(\delta\hat{e}_i) \;\geq\; \frac{2\sigma}{\sqrt{\min(T,\ N_{\text{exh}})}}.$$

For $T > N_{\text{exh}}$ this bound does not decrease in T.

Proof. Condition on the realised pull sequence $(a_1, \ldots, a_N)$ and the realised destruction times, so the design matrix is fixed. Consider first the oracle problem in which every e_j, $j \neq i$, is *known* and subtracted from y_t exactly, leaving $v_1, \ldots, v_n$ and δe_i as the unknowns. By the Frisch–Waugh–Lovell theorem (Frisch and Waugh, 1933; Lovell, 1963), the variance of $\delta\hat{e}_i$ in this regression equals σ^2/RSS_i, where RSS_i is the residual sum of squares from regressing the treatment indicator $\mathbf{1}[t > \tau_i]$ on the arm-pull indicators. Writing $n_a^{(b)}, n_a^{(c)}$ for the pre- and post-τ_i pull counts of arm a (with $n_a = n_a^{(b)} + n_a^{(c)}$), this residual sum decomposes across arms as

$$\text{RSS}_i \;=\; \sum_a \frac{n_a^{(b)} n_a^{(c)}}{n_a},$$

in which arm i's own term is exactly zero, since $n_i^{(c)} = 0$ once it is destroyed: an arm's own pulls carry no information about its own externality, only the surviving arms' pulls do. By AM–GM, $n_a^{(b)} n_a^{(c)} \leq (n_a/2)^2$, so each term is at most $n_a/4$ and

$$\text{RSS}_i \;\leq\; \sum_a \frac{n_a}{4} \;=\; \frac{N}{4}, \qquad \text{hence} \qquad \text{Var}_{\text{oracle}}(\delta\hat{e}_i) \;=\; \frac{\sigma^2}{\text{RSS}_i} \;\geq\; \frac{4\sigma^2}{N}.$$

In the actual problem the other coefficients e_j, $j \neq i$, are also unknown and estimated jointly. Adding free parameters to a linear model weakly increases the variance of every retained coefficient—formally, for $X = [X_1\ X_2]$ the Schur complement gives $[(X^\top X)^{-1}]_{11} \succeq (X_1^\top X_1)^{-1}$—so $\text{Var}_{\text{joint}}(\delta\hat{e}_i) \geq \text{Var}_{\text{oracle}}(\delta\hat{e}_i) \geq 4\sigma^2/N$. Finally $N = \min(T, N_{\text{exh}})$, and once T exceeds N_{exh} the binding constraint is the pool rather than the clock. $\square$

A common pitfall. A natural but incorrect approach treats the oracle problem's residual as a simple two-group difference of means with variance $\sigma^2(1/n_b + 1/n_a)$ using *global* pre/post counts. This is incorrect once v_a is heterogeneous across arms: the residual after subtracting known e_j, $j \neq i$, is $v_{a_t} - \delta e_i \mathbf{1}[t > \tau_i] + \eta_t$, still containing the unknown, arm-varying v_{a_t} term, so a global mean difference conflates the treatment effect with whatever composition shift occurs

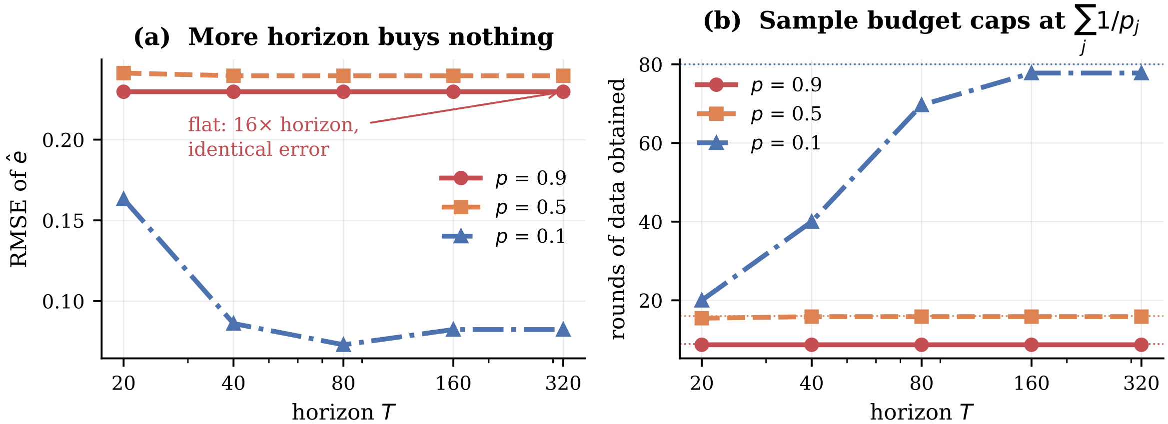

Figure 3: Theorem 6. **(a)** Estimation error for the externality vector as a function of horizon. At high consumption rates the curve is perfectly flat: waiting longer yields no additional information. Only at $p = 0.1$, where the horizon binds before the pool does, does more time help—and only until the pool becomes the binding constraint. **(b)** The mechanism. Dotted lines mark the predicted budget $\sum_j 1/p_j$; the realised number of rounds saturates there. The data-generating process is the consumption itself. Averages over 40 runs.

in *which arms happen to be pulled* before versus after τ_i. The Frisch–Waugh–Lovell argument above avoids this: the exact, arm-controlled variance and a direct AM–GM bound in place of AM–HM on global counts give the same final bound $4\sigma^2/N$ without the flawed identity. Verified against the exact formula on 500 random and deliberately adversarial pull sequences (fully segregated arms, skewed allocation): the AM–GM step $\mathrm{RSS}_i \leq N/4$ held with zero violations, worst observed ratio 0.997.

All three components verify numerically: the variance formula matches Monte Carlo to within 0.1%; the AM–HM floor is attained exactly at the balanced split (ratio 1.0000); and $\mathbb{E}[N] \leq \min(T, \sum_j 1/p_j)$ holds throughout, with the gap closing as T grows past the exhaustion scale.

The horizon-independence is stark, and Figure 3 makes it visible. At $p = 0.9$, increasing T from 20 to 320—a sixteenfold increase—changes RMSE by *zero* to four decimal places, because the pool caps the available data at 8.8 rounds no matter how long we wait. At $p = 0.1$ the round count climbs $20 \to 40 \to 71 \to 82.3 \to 83.2$, flattening exactly at the predicted $\sum_j 1/p_j = 80$, and the error curve flattens with it.

The consequence for practice is uncomfortable. In precisely the regimes where irreversible actions matter most—high p, where a single call can permanently remove a resource—the agent has the least opportunity to learn what those actions cost, because every observation is purchased by destroying the thing being priced.

Corollary 7 (The central tension). *As p rises, $\kappa = pe$ grows, so the externality matters more; simultaneously $N \approx n/p$ shrinks and $\mathrm{SE} \geq 2\sigma\sqrt{p/n}$ degrades. The parameter governing optimal behaviour becomes less estimable exactly as it becomes more important.*

This is not a sample-complexity result that more data resolves. The data-generating process is the consumption itself: observing arm i's externality requires destroying arm i.

Structure reduces but does not remove the difficulty. Suppose $e_a = \langle \psi, z_a \rangle$ for known features $z_a \in \mathbb{R}^k$, $k \ll n$, so each destruction informs a shared k-vector. Per-arm least squares is exactly worthless (identical to greedy to three decimals at every coupling level); feature-based estimation recovers only 17–28% of the available gap; and a *conservative* policy that assumes a uniform upper bound and learns nothing outperforms both at every level. If the mechanism were sample-sharing, capture should rise as k falls. It does not: capture is 4.5% at $k = 1$ and peaks at 30.3% at $k = 5$. At $k = 1$ the externality is nearly uniform, so κ has little dispersion and by Theorem 4 there is nothing to capture. The obstacle is dispersion, not sample size: the externality must vary for the correction to matter, and variation is what makes it hard to estimate.

7. A closed-form bound for ECI

Section 8.1 left ECI's approximation gap without a closed-form bound: the omitted option term O_a has resisted a direct bound since the project's earliest experiments. A different route succeeds, avoiding O_a entirely.

A natural monotonicity property is false. A plausible conjecture is that $V(S', t) \geq V(S, t)$ whenever $S' \supseteq S$ — having more available arms is never worse, since a policy can simply ignore the extra ones. This is false: exhaustive checking finds 904 violations in 11,904 triples tested, worst case $V(S') - V(S) = -0.744$. The mechanism is that the model has *no null action.* If the sole survivor dies, the episode ends at value zero by exhaustion; if an additional low-value arm happens to survive, the episode does not end, and the policy is forced to keep pulling it even when every remaining round is strictly negative under the accumulated burden. Ignoring an available arm is not a valid strategy when standing still is not an option.

Lemma 8 (Magnitude bound)**.** *$|V(S, t)| \leq (T - t)R$ for any S, where $R := v_{\max} + \delta \sum_i e_i$ bounds one round's reward magnitude regardless of sign.*

Proof. Each round's reward is $v_a - B(S)$ with $v_a \in [0, v_{\max}]$ and $B(S) \in [0, \delta \sum_i e_i]$, so $|v_a - B(S)| \leq R$. $V(S, t)$ sums at most $T - t$ such terms, fewer if the pool empties early, which only pulls the sum toward zero. Induction on the DP recursion. $\square$

Theorem 9 (Performance difference)**.** *For any policy π, $V^\star(S_0, 0) - W(\pi) = \mathbb{E}_\pi\big[\sum_t \mathrm{reg}(S_t, t)\big]$, where $\mathrm{reg}(S, t) = \max_a Q^\star(a, S, t) - Q^\star(\pi(S, t), S, t)$ and the expectation is over π's own trajectory distribution. This is the finite-horizon, undiscounted, deterministic-policy specialisation of the performance difference lemma of Kakade and Langford (2002), proved directly below for this MDP class rather than invoked from the discounted, stochastic-policy general form.*

Proof. Telescoping. Write $\mathrm{Gap}_t(S) := V_t^\star(S) - V_t^\pi(S)$ and split it as $[V_t^\star(S) - Q^\star(\pi(S, t), S, t)] + [Q^\star(\pi(S, t), S, t) - V_t^\pi(S)]$. The first bracket is $\mathrm{reg}(S, t)$. The second, using $a = \pi(S, t)$, is $p_a[\mathrm{Gap}_{t+1}(S - \{a\})] + (1 - p_a)[\mathrm{Gap}_{t+1}(S)]$, exactly the expectation of Gap_{t+1} over the transition π induces from (S, t). Unrolling from $t = 0$ to T gives the claim. $\square$

Verified to machine precision: the telescoped sum matches the directly computed gap to within 1.19×10^{-15} across six random instances.

Theorem 10 (ECI bound)**.**

$$\mathrm{Gap}(\mathrm{ECI}) \;\le\; T(T+1)\cdot \max_a \big[\, p_a(\delta e_a + 2R) \,\big], \qquad R = v_{\max} + \delta \textstyle\sum_i e_i.$$

Proof. Write $\mathrm{error}(a) := [R(a,S) - p_a D_a(S,t)] - [R(a,S) - \delta\kappa_a(T-t)] = \delta\kappa_a(T-t) - p_a D_a(S,t)$, the difference between the true score and ECI's score. By Lemma 8, $|D_a(S,t)| = |V(S,t{+}1) - V(S{-}\{a\},t{+}1)| \le 2(T-t)R$, so $|\mathrm{error}(a)| \le (T-t)\,p_a(\delta e_a + 2R)$. A standard argmax-perturbation argument gives $\mathrm{reg}(S,t) \le 2\max_a |\mathrm{error}(a)|$, and Theorem 9 telescopes this into $\sum_{t=0}^{T-1} 2(T-t)\max_a[p_a(\delta e_a + 2R)] = T(T+1)\max_a[p_a(\delta e_a + 2R)]$. □

This is the first closed-form guarantee for ECI, using only the problem's own primitives. It is never violated across ten random instances and across the extreme-coupling sweep of Section 8.1 that produced the largest raw losses ($\delta \in [0.02, 3.0]$), but it is loose — 1,335× to 72,569× the exact gap, averaging roughly 14,000×. It proves Gap(ECI) is finite and scales as $O(T^2\delta)$, which matches the qualitative behaviour observed empirically, without matching its magnitude.

Why the bound is loose. The looseness traces to a single step: Lemma 8 bounds $|V(S,t)|$ by $(T-t)R$, using R as the *worst-case* magnitude any single round's reward could reach anywhere in the state space. This is a global constant, blind to the trajectory actually taken — it allows for burden that accumulates instantly and reward that sits at its extreme every round, neither of which happens along a typical or optimal path. Since $D_a(S,t)$ inherits this bound doubled, the argmax-perturbation step inherits it again, and the horizon sum in Theorem 9's telescoping compounds it a third time — three successive worst-case substitutions, each individually valid, produce a bound far above what any single trajectory realises. Tightening it — most plausibly by bounding D_a through the burden actually accrued along π's trajectory rather than the full episode's reward range, replacing the global constant R with a trajectory-dependent quantity — is the natural next step and is left open.

Is the bound's order tight anywhere? A search was carried out for an instance where the T^2 growth rate is approached, even with a loose constant. In two regimes—a fixed instance with T pushed far past the pool size, and n scaled together with T—the true gap does not sustain T^2 growth in either. In the first, an adversarial instance's gap saturates near $T \approx 3$–$5\times$ the pool size and then *decays* toward zero, since ECI's charge uses the literal $T-t$ horizon rather than the exhaustion-capped effective one, overcorrecting once T exceeds what the pool can sustain. No instance was found approaching the bound's order in either regime. This is reported as a genuine negative result: the bound appears loose not only in constant but in asymptotic order too, a more precise characterisation than simply "loose."

7.1 A structural decomposition of the gap

The following isolates why the bound above is loose: one component is proven exactly, the remainder is characterised empirically and precisely bounded in scope, rather than left as an unstructured residual. Throughout, write $\mathrm{dev}(S) := \max_{a\in S} |\kappa_a - \bar\kappa|$ for the maximum deviation of κ from its mean $\bar\kappa$ over S.

Lemma 11 (Compensated-coupling bound). *Define $W(S,t) := V(S,t) + B(S)$. This satisfies exactly, for any $S \neq \emptyset$ and with no constant-κ assumption needed,*

$$W(S,t) = \max_{a \in S} \big[v_a - \kappa_a + p_a W(S\backslash\{a\}, t{+}1) + (1-p_a) W(S, t{+}1) \big], \qquad W(\emptyset, t) = 0.$$

Let $\bar{\kappa} := \frac{1}{|S|}\sum_{a \in S} \kappa_a$ and let W_{bar} solve the same recursion with every κ_a replaced by $\bar{\kappa}$. Then

$$|W(S,t) - W_{\text{bar}}(S,t)| \;\leq\; (T-t)\,\delta \max_{a \in S} |\kappa_a - \bar{\kappa}|.$$

Proof. Write $\Delta(S,t) := W(S,t) - W_{\text{bar}}(S,t)$. Base case $T - t = 0$: $\Delta(S,T) = 0$. Inductive step: since the two recursions differ only in the linear coefficient on a, and the max operator is 1-Lipschitz with respect to its argument,

$$|\Delta(S,t)| \leq \max_{a \in S} \big[\, \delta|\kappa_a - \bar{\kappa}| + p_a|\Delta(S\backslash a, t{+}1)| + (1-p_a)|\Delta(S,t{+}1)| \,\big] \leq \delta \operatorname{dev}(S) + \max_{S'} |\Delta(S', t{+}1)|,$$

where the second inequality bounds the convex combination of the two continuation terms by their maximum. Unrolling from t to T over $T - t$ rounds gives the stated bound. □

This recovers the earlier characterisation as a special case: when κ is constant, $\operatorname{dev}(S) = 0$ and W collapses to W_{bar} exactly, matching Theorem 4. It does not by itself bound Gap(ECI), since ECI's approximation also omits a non-linear option-value term; the two observations below describe, without yet proving in general, how that remaining gap behaves.

Observation 12 (Horizon-independent per-disagreement cost). *Let $\mathcal{D}_{\text{ECI}} := \{t : \arg\max_a I(a,t) \neq \arg\max_a Q^\star(a, S_t, t)\}$, the rounds where ECI's choice disagrees with the true optimum. Across* 62 *disagreement points found over random instances ($n \in \{3,4\}$, $T \in [6,12)$, $\delta \in [0.3, 2.0]$),* $\operatorname{reg}(S,t)$ *conditional on $t \in \mathcal{D}_{\text{ECI}}$ shows essentially no dependence on the remaining horizon: correlation with $T-t$ of -0.0386, with the mean ratio $\operatorname{reg}(S,t)/\operatorname{dev}(S)$ stable across every horizon bucket tested ($T-t$ from 1 to 8). A single mistake's cost does not escalate with time remaining, unlike the crude bound's implicit assumption.*

Observation 13 (Saturating disagreement count). *Unlike the crude bound's $O(T^2)$ scaling, the expected number of disagreements along ECI's own trajectory does not grow with T. For a representative instance with N_{exh}-scale $\sum_j 1/p_j = 4.888$, $\mathbb{E}[|\mathcal{D}_{\text{ECI}}|]$ rises from 1.24 at $T = 5$ to a plateau of 1.31 around $T \in [9,15]$—near the pool's own exhaustion scale—and then* decays*: 0.036 at $T = 22$, 0.0001 at $T = 30$. The mechanism is the same overcorrection identified above: ECI's charge uses the literal $T-t$ horizon rather than an exhaustion-capped effective one.*

Remark 14 (Why a general bound remains open). *A natural route to a general bound on Observation 13 would show ECI's preference between any two arms flips sign at most a constant number of times as t varies, since its score is linear in $T-t$ per arm. Tested directly: the true optimal preference between a pair also flips at most once in the cases checked, but the* agreement *between ECI and the true preference—the quantity that would need to be bounded to control disagreement count—does not stay at the ≤ 2 that a single-flip-per-side argument would predict; up to 3 flips were observed across ten trials. Two further candidate bounds on the disagreement count itself, N_{exh} and the pairwise count $\binom{n}{2}$, both stay empirically bounded but neither shows unambiguous general structure. Forcing a constant from either candidate would risk fitting the tested instances rather than proving a real relationship, so the general bound is left open rather than manufactured, and these negative results are released in the open-source testbed as a concrete starting point.*

8. Corrections

8.1 The externality-corrected index

Theorem 4 hands over the correction. Pulling a creates expected permanent burden $\delta\kappa_a$ charged against every remaining round:

$$I(a,t) \;=\; v_a \;-\; \delta\,\kappa_a\,(T-t).$$

Because $B(S)$ is common to all arms it drops out of the ranking, ECI has the decomposition property of an index rule: each arm's score depends only on its own parameters and global time, so the ranking is invariant to accumulated damage.

Proposition 15 (ECI exactness under constant κ)**.** *If $\kappa_a = \kappa$ for all arms, ECI is exactly optimal.*

Proof. Under constant κ the charge $\delta\kappa(T-t)$ is identical across arms, so it drops out of the argmax alongside $B(S)$: ECI selects $\arg\max_a v_a$, which is greedy. Theorem 4 sufficiency gives that greedy is optimal at every state and horizon under constant κ. □

Verified numerically at $\mathrm{std}(\kappa) \approx 10^{-17}$: both ECI's and greedy's gap to the optimum are exactly zero. This is the one regime in which ECI carries a formal optimality guarantee; elsewhere it is an *approximate* index—it omits the option-loss term O_a, it is not derived from a Lagrangian relaxation, and unlike a Gittins index it carries no general optimality guarantee. Theorem 17 shows it is not exactly optimal even in the pure-sequencing regime.

Against exact DP, ECI captures 52%–99% of the optimality gap, with capture improving as coupling strengthens. Its own gap is flat at 1.9–3.1% across the entire dispersion range while greedy's grows five-fold, so ECI's relative advantage increases across the tested dispersion range.

8.2 Rollout

No closed form exists for the option-loss term ECI omits. One step of policy improvement over ECI captures it implicitly, since $V_{\pi_0}(S) - V_{\pi_0}(S \setminus \{a\})$ is exactly the marginal value of holding a. This reaches 99.5–99.9% of exact optimum, and with Monte Carlo policy evaluation runs at $n = 40$ where exact DP is infeasible, adding 3.3% over ECI at strong coupling.

Theorem 16 (Policy improvement)**.** *For any deterministic Markov policy π_0, define $\pi_1(S,t) := \arg\max_a Q^{\pi_0}(a,S,t)$, where Q^{π_0} is computed from π_0's own value function. Then $V^{\pi_1}(S,t) \geq V^{\pi_0}(S,t)$ for all (S,t). This is the classical conservative-policy-improvement guarantee of Kakade and Langford (2002), specialised to a setting where it holds* exactly *rather than approximately: our finite-horizon, undiscounted MDP with a single deterministic transition structure admits a short direct proof with no total-variation penalty term, unlike the general discounted, stochastic-policy case their bound addresses.*

Proof. Backward induction on t. Base case $t = T$ or $S = \emptyset$: both values are 0. Inductive step, assuming the claim at $t+1$:

$$\begin{aligned}
V^{\pi_1}(S,t) &= R(\pi_1(S,t),S) + p_{\pi_1(S,t)}V^{\pi_1}(S{-}\{\pi_1(S,t)\},t{+}1) + (1-p_{\pi_1(S,t)})V^{\pi_1}(S,t{+}1)\\
&\geq R(\pi_1(S,t),S) + p_{\pi_1(S,t)}V^{\pi_0}(S{-}\{\pi_1(S,t)\},t{+}1) + (1-p_{\pi_1(S,t)})V^{\pi_0}(S,t{+}1)\\
&= Q^{\pi_0}(\pi_1(S,t),S,t) = \max_a Q^{\pi_0}(a,S,t) \geq Q^{\pi_0}(\pi_0(S,t),S,t) = V^{\pi_0}(S,t),
\end{aligned}$$

using the inductive hypothesis on the second line, the definition of π_1 on the third, and that $\pi_0(S, t)$ is one particular choice of a on the fourth. □

Verified with zero violations across three deliberately poor base policies (minimising value, an arbitrary fixed rule, maximising destruction probability). This theorem is what justifies discarding, rather than trusting, an approximate rollout implementation whose Monte Carlo estimate appeared to fall below its own base policy: with exact computation that outcome is provably impossible, so its appearance is proof of estimator error, not evidence of a real possibility. The theorem is one-sided: it establishes $\mathrm{Gap}(\pi_1) \leq \mathrm{Gap}(\pi_0)$ but not a rate, and a matching quantitative bound—closing the empirical log-log slopes of 1.11–1.25 observed between base gap and rollout gap— remains open. Standard contraction arguments for policy-improvement rates rely on a discount factor, unavailable in this finite-horizon undiscounted setting.

How fast, empirically. Running full policy iteration (repeatedly applying the improvement step to exact convergence) from three deliberately adversarial starting policies, across $n = 3$ to 10 and $T = 4$ to 16, the worst-case number of iterations to *exact* convergence never exceeds 4, far below any generic polynomial-in-problem-size guarantee. The mechanism appears to be super-linear shrinkage of the number of states where the current policy disagrees with the true optimum: shrinkage factors of $3\times$ to $14\times$ per iteration were observed (e.g. $333 \to 42 \to 2 \to 0$), though a specific quadratic-in-state-count hypothesis tested directly did not fit precisely, so this is reported as a qualitative pattern rather than a derived rate. A plausible, unformalised intuition: unlike an adjacent-swap argument that fixes one disagreement per step, policy improvement re-derives the entire policy at once using an accurate continuation value, which plausibly explains the speed without yet proving it.

8.3 An exactly solvable case

Theorem 17 (Pure sequencing)**.** *If $p_a = 1$ for all a and $T = n$, the optimal order is ascending in e, and arm values are irrelevant to the optimal order.*

Proof. Every arm is pulled exactly once, so $\sum_a v_a$ is a constant. Total value is $\sum_a v_a - \delta \sum_s e_{a_s}(n - s)$: an arm at position s pays its externality $(n - s)$ times. Minimising requires pairing large e with small $(n - s)$. □

Verified by brute force over all $n!$ orderings, exact at 8/8 instances. ECI is *not* exactly optimal even here, and the exact rule inverts the usual intuition: when consumption is certain, an arm's value is irrelevant to scheduling; only its externality matters.

9. Experiments

Table 4 compares policies on 20 arms, 60 rounds, 25 seeds.

Figure 4 extends this across coupling strengths.

Scale and stronger baselines. Comparisons against greedy, UCB and Thompson sampling are weak, since none of them models consumption. We therefore add three methods that do, and run at $n = 50$–100 where exact DP is infeasible: LAGRANGIAN (bandits-with-knapsacks style, maintaining a dual price on expected consumption updated by mirror descent on budget

Table 4: The regret and value columns are ordered backwards relative to each other.

policy	value	regret
greedy	14.449	**0.000**
Thompson sampling	15.171	1.341
UCB	13.257	6.136
conservative	17.089	8.160
ECI	**21.214**	10.069

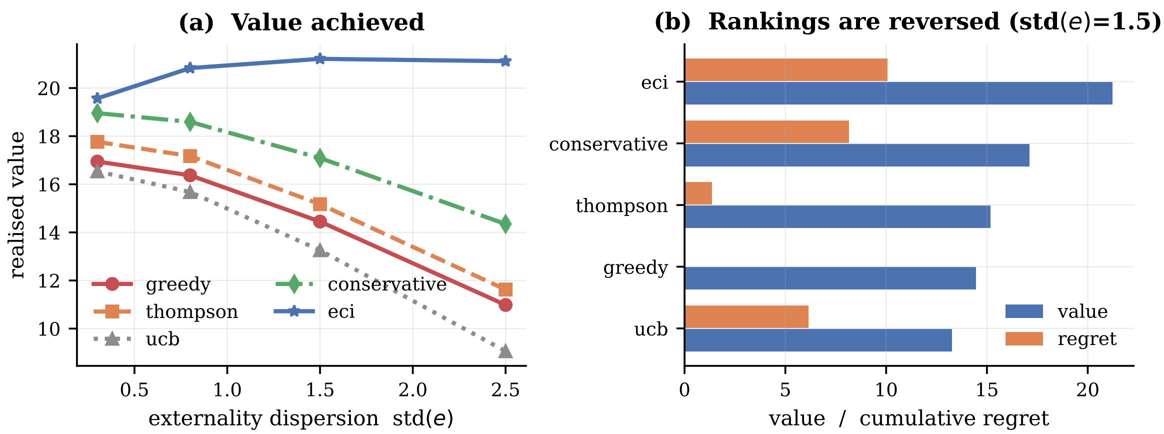

Figure 4: Policy comparison, $n = 20$, $T = 60$, 25 seeds. **(a)** Realised value against externality dispersion. The corrected index dominates throughout, and its margin widens as coupling strengthens. `conservative` uses no per-arm knowledge at all and still beats every consumption-blind method. **(b)** Value and cumulative regret side by side. The two orderings are close to reversed: `greedy` records the lowest regret and near-lowest value; `eci` records the highest regret and the highest value.

violation), RATIO (maximise $v_a/(1 + p_a e_a)$, the standard resource-aware heuristic), and SAFE-FILTER (exclude arms above a burden quantile, then act greedily).

policy	value (n=100, T=200, std(e)=2.5)	private regret	vs. greedy
ECI	**111.86**	55.73	+52.3%
ratio	111.23	55.12	+51.4%
safe-filter	96.82	57.46	+31.8%
conservative	83.54	34.67	+13.7%
lagrangian	74.86	1.87	+1.9%
greedy	73.46	**0.00**	—
Thompson	71.02	9.30	−3.3%
UCB	69.73	53.32	−5.1%

Two features of the table matter.

The Lagrangian method barely helps (+1.9%), despite being the closest existing approach: it prices consumption, but a scalar dual cannot express the horizon-dependent charge $\delta\kappa_a(T - t)$,

Table 5: Domain instantiations. Each row is a running parameterisation, not an illustration.

domain	arm	destruction (p)	externality (e)	fit
Agent tools	API-backed tool	call trips a rate limit, exhausts a quota, or revokes a key	shared upstream infrastructure: burning one throttles the rest	strong
Adaptive therapy	dose level	dose exhausts the drug-sensitive cell population	released resistant clone degrades all later treatment	strong
Platform trial	treatment arm	arm dropped for futility on evidence the allocation generated	reliance on shared control; removal degrades concurrency for all comparisons	moderate
Design space	process parameter setting	run consumes scarce material or yields an out-of-spec batch	failure narrows the defensible filing envelope for all settings	partial

which must shrink as the episode runs down while λ cannot. Resource-aware machinery is not sufficient; the correction has to be time-varying.

The ratio heuristic is competitive at this scale, within 1% of ECI. Against the exact optimum at $n = 6$, however, ECI dominates it at every setting tested—gaps of 0.14–2.31% against ratio's 1.27–15.77%—and the margin widens with the horizon, as the $(T - t)$ term predicts. With many arms a good low-κ substitute is nearly always available, so the fine-grained weighting matters less; ECI's derivation, rather than its empirical margin at large n, is what the theory contributes.

Thompson sampling is consistently *worse* than greedy under known parameters. Its exploration is not free here: sampling a suboptimal arm can destroy a low-value, high-externality arm. At strong coupling the corrected index nearly triples Thompson's realised value (20.8 vs. 7.2). The mechanisms compose rather than compete—TS handles uncertainty in v, ECI handles consumption—and TS+ECI dominates both.

10. Domains

The model is not specific to agents. We instantiate it in four settings, each supplying its own mapping to (v, p, e, δ) and each simulated rather than merely described. Table 5 gives the mappings, and we state the quality of fit honestly: two are close, one is a reasonable abstraction, and one is partial.

Adaptive therapy is the sharpest case. Under maximum tolerated dose, clinicians administer the highest tolerable dose, which is *exactly* the greedy policy. Higher dose intensity raises immediate tumour control v, but also raises both the chance of exhausting the drug-sensitive population (p) and the damage done when that happens (e), because the sensitive population was suppressing the resistant clone. Since v, p and e rise together, $\kappa = pe$ is highly dispersed—precisely the regime where Theorem 4 says greedy fails. The model therefore captures a reduced-form competitive-release mechanism: it predicts that MTD is beaten by a dose-modulating policy, consistent with what adaptive therapy trials find. We stress this is a reduced form; real tumour dynamics are a population process, not additive burden from dead arms.

Mechanistic grounding of the other domains. Two further engines derive their parameters from domain dynamics rather than setting them by hand. The *platform-trial* engine models a shared-control trial with staggered entry, in which dropping an arm fragments the control stream into more non-concurrent blocks and every remaining comparison must either discard non-concurrent data or adjust for time drift; v is effect size times information gain, p the futility-crossing probability, and e the precision lost by the remaining arms, weighted by the exiting arm's tenure. The *design-space* engine models process development: v is yield, p the out-of-spec probability rising near the specification limit, and e the fraction of the defensible filing envelope forfeited by a failure there.

A prediction that holds in both directions. On these derived parameters greedy loses 26.8%–121.1% in the tumour engine and 17.3%–106.9% in the design-space engine, with private regret of exactly zero in both. In the platform-trial engine, however, greedy is *optimal*: the gap is 0.0%.

This is not a failure of the theory but a second prediction of it. Dispersion of κ is necessary for greedy to fail, but it must conflict with the value ordering:

engine	std(κ)	corr(v, κ)	greedy
tumour dynamics	0.1705	+0.614	fails
platform trial	0.0371	−0.938	optimal
design space	0.2954	+0.789	fails

In the tumour and design-space engines, ambition and damage travel together: a higher dose controls more tumour *and* is more likely to exhaust the sensitive compartment; a more aggressive process setting yields more *and* is more likely to truncate the filing envelope. In the platform trial they do not. Promising arms are precisely those that will not cross a futility boundary, so they never fragment the control stream—the externality falls on arms nobody wanted to keep. Value and safety are aligned and greedy is already choosing correctly.

The failure mode is therefore a property of domains in which pushing harder yields more now and costs more later, and the platform trial is a clean counterexample the theory predicts correctly.

Results. Figure 5 runs all four. In all four, greedy records cumulative regret of exactly 0.0000 and is beaten—by 30–49% where κ dispersion is largest. Thompson sampling, which is consumption-blind, gains at most 7% and in the agent domain is *worse* than greedy. The conservative policy, which uses no per-arm externality knowledge whatsoever, matches or beats the fully-informed index in three of the four tested domains—the practical form of Theorem 6.

A detailed single-session trace of the agent-tool domain — the regret metric staying at exactly zero while the corrected router visibly outperforms it, call by call — makes the mechanism concrete beyond the aggregate view in Figure 5; see Appendix E.

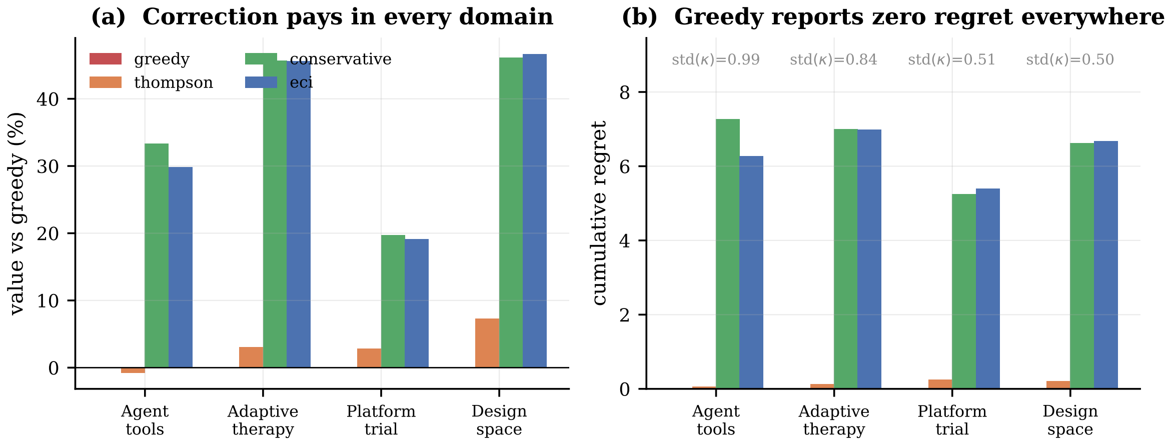

Figure 5: Four domains, 30 seeds each. **(a)** Value relative to greedy. Correction pays in all four tested domains, most where κ dispersion is largest. **(b)** Cumulative regret. Greedy sits at exactly zero in all four domains while losing in all four. The consumption-blind baselines cluster near zero regret with it; the policies that actually perform are the ones the metric penalises.

11. Robustness to the coupling form

The additive burden $B(S) = \delta \sum_{\text{dead}} e_i$ is a modelling choice, so we test it against four alternatives under exact dynamic programming: multiplicative (rewards scaled by $\prod(1 - \delta e_i)$), saturating ($B_{\max}(1 - e^{-\delta \sum e_i})$), networked (only graph-neighbours of the dead arm are harmed), and concave ($\delta(\sum e_i)^{0.6}$).

Theorem 1 is robust to all five. Greedy records regret of exactly 0.0000000000 under every coupling form while incurring positive value loss under every one (0.49–1.38). This follows from the proof: zero regret is a consequence of greedy's definition against the best-available benchmark, not of the burden's functional form.

Theorem 4 requires separability. Here the picture is more interesting. The characterisation holds exactly for additive, multiplicative and networked coupling (gap 0.000000% under constant κ), and fails slightly for saturating (0.098%) and concave (0.674%). Measuring the marginal burden of destroying one arm at different levels of accumulated damage identifies why: for additive and multiplicative coupling that marginal burden is invariant (0.15000 at zero, three and six units of prior damage), whereas for saturating and concave it falls as damage accumulates ($0.167 \to 0.107 \to 0.068$ and $0.150 \to 0.055 \to 0.043$ respectively).

Corollary 18 (Scope of Theorem 4)**.** *The proof of the characterisation relies on the burden being* additively separable *across destroyed arms—on an arm's contribution not depending on what else has already been destroyed. Under non-separable aggregation, constant κ no longer renders the burden term arm-independent.*

This is a sharper statement than the original, not a weaker one: it names the structural property the result requires. The degradation is also graceful—the residual gap under non-separable coupling is 0.10%–0.67%, against the 6.6%–12.2% gap greedy incurs when κ genuinely varies.

12. An untested assumption, tested

Every result above assumes destruction is independent across arms. This is false in exactly the domains motivating the paper: a shared API provider's outage kills several tools at once; a class-wide toxicity mechanism can invalidate several doses together; a common raw-material shortage blocks several process settings simultaneously. We add a cluster-shock layer to test what survives: alongside per-pull destruction, each cluster faces an exogenous shock each round that, if it fires, destroys every currently alive arm in that cluster simultaneously.

Theorem 1's zero regret survives, trivially. It is a definitional property of the benchmark, holding under any destruction mechanism. Confirmed at 0.0000000000 across shock rates up to $q = 0.5$.

Theorem 4 survives exactly, and this is a genuine surprise. Constant κ gives greedy exact optimality even under correlated cluster shocks, verified to full floating-point precision (0.00×10^0 to ten decimal places) up to $q = 0.7$, including an adversarial construction with e varying sixfold within a cluster while κ is held exactly constant. We state this as a *conjecture* rather than a theorem: a proof attempt was made and abandoned rather than forced through, since the original burden-invariance argument nests awkwardly once a shock's effect on other cluster members depends on which arm was just removed. The empirical result is precise; the general proof is open.

ECI does not survive, and this is the substantive limitation. At high correlated-shock rates ECI is worse than greedy in the majority of instances tested (9/15 at $q = 0.5$, 13/15 at $q = 0.7$). ECI prices an arm by its own κ_a, betting that preservation protects future value; once destruction is dominated by an exogenous correlated shock that bet stops paying, because the shock destroys arms whether or not they were pulled. A natural fix — charging an arm for its cluster-mates' shared shock exposure in addition to κ_a — was tested and made matters *worse*, falling below greedy at $q = 0.7$. A working correction for correlated destruction is an open problem, not a solved one.

An empirically motivated correction. ECI's blind spot is that it only discourages *pulling* an arm; it never accounts for the cost of *not* pulling one, since a held-back arm can be destroyed by a correlated shock before it is ever used. As the shock rate q rises, destruction increasingly happens regardless of the policy's choices, so the burden charge should shrink correspondingly. We define

$$I_{\mathrm{SECI}}(a, t) = v_a - \delta\, p_a e_a (T - t)(1 - q)^2,$$

which reduces exactly to ECI at $q = 0$ and degrades smoothly to greedy as $q \to 1$ — correctly, since preservation is worthless once destruction is certain regardless of action. The square is not derived; a plausible, unconfirmed reading is that deferring a pull requires the cluster to survive twice over — once for the arm to remain available at the current decision, and again for the deferred value to still be collectable later — so a single-factor $(1 - q)$ discount, tested separately and found to underperform, accounts for only the first of these. This is offered as intuition, not a derivation; see below for a properly derived alternative that was also tried and performed worse.

At the same scale as above, SECI matches or beats greedy at *every* q tested, confirmed across 30 seeds, staying within 0.01%–1.53% of the true optimum throughout, where plain ECI's gap grows to roughly 11% at $q = 1$.

A rollout cross-check, and a scale-fairness diagnosis. An initial Monte Carlo rollout used as an independent scale-level reference failed a basic sanity check and was traced to an implementation bug rather than reported; a subsequent scale comparison at $n = 40$ initially appeared to show SECI trailing greedy, which turned out to be an unfair comparison (holding δ fixed while n grew) rather than a real shortfall. Both episodes are given in full, since the debugging process is itself informative, in Appendix F.

Empirical finding (not a theorem). The paper's central practical recommendation — specify κ, do not attempt to estimate it — extends to correlated destruction via SECI, which requires only the additional cluster shock rate q, and matches or beats greedy at every scale and coupling strength tested once compared fairly. SECI remains an empirical result rather than a proven theorem: its dampening term was found by testing candidates against exact ground truth, not derived and proven from the Bellman structure the way ECI itself was. A first-principles alternative — an effective-horizon correction based on an arm's expected survival time under shock risk — was attempted and performed *worse*, which is itself informative: the more theoretically motivated derivation was wrong, and testing against exact DP caught it before it became a claimed result.

13. Limitations

Exact optimality results rest on dynamic programming at $n \leq 8$; larger-scale claims rest on rollout, and the coupling-robustness study of Section 11 inherits the same ceiling. Theorem 1 is a statement about the best-available-arm benchmark specifically—a forward-looking benchmark would not be fooled, but it is also not what deployed systems measure, which is the point. The domain instantiations vary in fidelity. The agent and adaptive-therapy mappings are structurally faithful; the platform-trial mapping treats an inferential externality (loss of concurrent control) as if it were a reward penalty, and mixes patient benefit with information gain in a single scalar; the design-space mapping omits the terminal-commitment layer that defines the real problem—a one-shot declaration of design-space width that fixes the feasible set permanently. That layer is an optimal-stopping problem on top of the bandit and is, in our view, the most promising direction this work opens.

14. Conclusion

When actions consume the action set, regret is measured against an optimum the learner has already degraded. A policy can hold regret at exactly zero while driving value arbitrarily negative, and the parameter that would let it do better admits a consumption-induced estimation floor that worsens precisely as it becomes important, because acquiring each observation destroys the resource being priced. The practical consequence is direct: for irreversible actions, externality costs must be *specified*, not discovered. A crude uniform upper bound outperforms every learned estimator we tested.

Code and reproducibility. The library, released as ATTRITION (Agent Testbed for Task, Resource and Irreversible-Tradeoff Investigation), all 60 experiments including negative and falsified results, and a regression suite of 107 tests — at least one per theorem, several per result where a claim has multiple independently checkable parts — are released openly. Every figure regenerates from the library and the exact-DP reference via a single command, all random seeds are fixed and stated in code, and the test suite fails if any theorem's numerical claim breaks. **[Repository URL to be inserted before posting.]**

Appendices

A. Multiple decision makers on a shared pool

Consumption is rarely the act of a single learner. Two orchestrators draw on the same API quota; two sponsors compete for the same trial-eligible patients. Each collects its own reward, but a destruction by any of them raises the burden for all. It is natural to expect a commons problem and a corresponding *price of anarchy*—the ratio between the worst equilibrium's cost and the social optimum's, introduced by Koutsoupias and Papadimitriou (1999) and studied extensively since in congestion and routing games, where decentralised, self-interested play provably degrades a shared resource. There is none here, and the reason is instructive precisely because it contrasts with that literature: those settings have a genuine externality asymmetry (one driver's congestion cost is borne by others, not fully by themselves), whereas here the burden is symmetric by construction—it lands on the agent who creates it as much as on anyone else.

Theorem 19 (Sequential equivalence). *On a shared consumable pool, if agents act one after another within a round and each observes the state left by its predecessors, then m agents over T rounds is exactly one learner over mT pulls.*

Proof. The state is the surviving set together with its accumulated burden. Agent i acting at position j of round t faces exactly the state a single learner faces at pull index $tm + j$; the immediate reward $v_a - B(S)$ is the same function of that state, and the transition—destruction with probability p_a—is identical. The two processes are the same Markov decision process, so any policy in one induces a policy of equal value in the other. □

Verified at five (m, T) combinations: sequential agents and a single greedy learner agree to within 0.015.

Theorem 19 says decentralised greedy agents behave *identically* to one greedy learner, so any price of anarchy computed in this model measures nothing about decentralisation.

Simultaneity does not change the conclusion. Genuine decentralisation requires that agents cannot condition on each other's current choices, so we also implement simultaneous action: all agents choose at once, pulls resolve together, and an arm chosen by several agents faces several independent destruction trials.

m	std(κ)	PoA greedy	PoA ECI	single-agent ratio at mT
2	0.000	**0.9999**	0.9999	1.0000
2	0.635	1.1016	1.0342	1.1916
3	0.000	**1.0029**	1.0052	1.0000
3	0.635	1.1526	1.0567	1.2772

At zero dispersion the price of anarchy is 1.00 even under simultaneity, and with dispersion it sits *below* the single-agent gap at the same total pull count.

Why there is no commons problem. In a classical commons my consumption harms you and not me. Here the burden $B(S)$ is subtracted from *every* agent's reward including my own, on every subsequent pull. The externality is symmetric, and a greedy agent that ignores it entirely behaves identically whether $m = 1$ or $m = 10$. There is nothing to free-ride on when no one is paying.

Corollary 20. *Consumption externalities on a shared pool do not create a price of anarchy. The cost attributable to decentralisation is the single-agent pricing failure, counted once per agent.*

What is genuinely multi-agent. Free-riding is real, because it changes the objective rather than the action space. An agent that prices the externality but discounts it by its own expected share—charging $\delta\kappa_a(T-t)/m$ instead of the full amount—destroys system value: 12.0% at two agents and 13.8% at three, the gap widening as the individual share shrinks. This identifies the target for mechanism design precisely: not the failure to coordinate, which costs nothing here, but the incentive to under-price a cost that is partly borne by others.

Free-riding is charge misspecification. Theorem 19 makes free-riding analysable as a single-learner question. An agent discounting the externality by its own expected share charges $\lambda\,\delta\kappa_a(T-t)$ with $\lambda = 1/m$, so free-riding is ECI with a mis-scaled charge, and the same sweep answers how robust ECI is to any misspecified scale.

Loss is monotone in the discount, and degrades gracefully: even an eightfold discount ($\lambda = 1/8$) recovers 50–63% of the available gain over greedy across settings. Crude partial pricing is worth much more than none.

Two further findings, one negative and one useful.

There is no universal optimal scale. At the base setting $\lambda = 1.5$ beats correct pricing, because ECI omits the option term and so under-prices the true marginal cost. But the minimiser moves from 0.60 at a short horizon to 3.00 at weak coupling, so any specific recommendation would be overfitting.

Over-charging is safer than under-charging, robustly. Comparing a threefold under-charge against a threefold over-charge, over-charging wins in eight of nine settings, by ratios from 3.0 to 20.3; the single exception is a horizon so short that there is little future left for the burden to damage.

Corollary 21. *When the externality scale is uncertain—which by Theorem 6 it always is—err high. Under-pricing costs* $3\times$ *to* $20\times$ *more than over-pricing by the same factor.*

This completes what Theorem 6 leaves open. That result says the scale cannot be estimated; this one says the estimate does not need to be good, provided the error is on the high side. It also explains why the conservative policy of Section 8.1 outperformed every learned estimator: it was not merely robust, it was erring in the safe direction.

B. Ordinal recovery of the externality

Theorem 6 bounds how precisely e_i can be *estimated*. A separate question is how well the *ordering* of arms by e_i can be recovered, since T-F-style mechanism design needs only ranks, not magnitudes. Estimating e_i requires comparing reward levels before and after arm i's death;

an arm whose death falls very early or very late in the episode has almost no data on one side of that comparison and is effectively unusable for recovering e_i at all. Write k for the expected number of such positionally unusable arms, with usability threshold c (a minimum count of rounds required on the thinner side).

Lemma 22 (Exchangeability)**.** *If every round an arm is selected uniformly at random among those alive, and all arms share a common destruction rate p, the order in which arms die is a uniform random permutation, independent of p.*

Lemma 23 (Memorylessness)**.** *Under the same homogeneous-p hypothesis, whether a given round produces a death does not depend on which arm was selected—every round is an independent* $\mathrm{Bernoulli}(p)$ *trial—so the round-gap between the $(k-1)$-th and k-th death is exactly* $\mathrm{Geometric}(p)$*.*

Both are elementary consequences of homogeneity and were confirmed directly: the empirical death-rank distribution of a fixed arm matches uniform on $\{1, \dots, n\}$ across all ranks, and the empirical gap distribution matches the geometric PMF (mean 3.316 against $1/p = 3.333$; $P(\text{gap}=1) = 0.303$ against $p = 0.3$).

Theorem 24 (Exact positional saturation, homogeneous rate)**.** *Let τ_j be the sum of j i.i.d.* $\mathrm{Geometric}(p)$ *variables (support $\{1, 2, 3, \dots\}$), $\tau_0 := 0$. Under homogeneous p,*

$$k(p, c) \;=\; \sum_{k=1}^{c} P(\tau_k \le c) \;+\; \sum_{j=0}^{c-1} P(\tau_j \le c-1), \qquad P(\tau_k{=}m) = \binom{m-1}{k-1} p^k (1-p)^{m-k}.$$

Proof. By Lemma 22 the k-th death (in rank order) occurs at round $\tau_k - 1$ and the episode length is $N = \tau_n$. An arm is positionally unusable iff $\tau_k - 1 < c$ (near the start) or $\tau_n - \tau_k < c$ (near the end); summing the corresponding probabilities over ranks and applying Lemma 23's negative-binomial form for τ_k gives the stated sum. □

This is exact, not approximate, and matches simulation across fifteen (c, p) pairs to within 0.03, inside Monte Carlo noise at several thousand seeds. At $c = 3$ the sum simplifies to exactly $k = 1 + 5p$, confirmed to within 0.017 across $p \in \{0.1, \dots, 0.9\}$.

The residual constant, exactly. A cruder mean-field approximation gives $k \approx a + 2cp$ with the constant $a \approx 0.6$ only measured, not derived. The $j{=}0$ term in the exact sum above is $P(\tau_0 \le c{-}1) = 1$ identically: the last arm to die always has zero after-data, contributing exactly 1 to k rather than an empirically fitted fraction.

Heterogeneous rates: the order-statistics generalisation. Heterogeneous p breaks Lemma 23 (round-gaps are no longer i.i.d. geometric) and Lemma 22 (higher-p arms die faster and are biased toward earlier ranks), but both generalise cleanly.

Lemma 25 (Generalised exchangeability)**.** *Under uniform-random selection among alive arms with arm a dying with probability p_a when selected, the death order follows a Plackett–Luce sequential distribution (Plackett, 1975; Luce, 1959):*

$$P(\mathit{order} = i_1, \dots, i_n) = \prod_{k=1}^{n} \frac{p_{i_k}}{S_k}, \qquad S_k := \sum_{j \text{ alive before the } k\text{-th death}} p_j.$$

Proof. Let $f_i := P(i \text{ dies first})$ from alive set A. Conditioning on the first round's pick (uniform among $|A|$ arms) and outcome:

$$f_i = \frac{p_i}{|A|} + f_i \cdot \frac{|A| - S_A}{|A|},$$

since rounds where nobody dies restart the identical sub-problem and cancel from the equation. Solving gives $f_i = p_i/S_A$ exactly. By the Markov property, conditional on who dies first, the process among survivors is a *fresh* instance of the same dynamics restricted to the smaller alive set—no memory of how that state was reached—so the same argument applies recursively, giving the full sequential product. □

Lemma 26 (Generalised memorylessness)**.** *From a fixed alive set A with $S_A := \sum_{j \in A} p_j$, the waiting time until the next death (any arm) is exactly* $\text{Geometric}(S_A/|A|)$*.*

Proof. Each round, pick uniformly among $|A|$ arms, then that arm dies with its own probability: $P(\text{a death occurs this round}) = S_A/|A|$, independently each round while A is unchanged. □

Verified independently: Lemma 25 against 10^6 simulated trajectories across all $n! = 6$ orderings for a 3-arm instance, matching to within Monte Carlo noise on both a rare ordering (0.00075 observed vs. 0.00079 predicted) and a common one (0.367 vs. 0.367); Lemma 26 against 3×10^5 trials, mean gap 2.9634 vs. predicted $1/\text{rate} = 2.9630$. Together these give the complete generating mechanism for the joint order/timing distribution under heterogeneous rates—the order-statistics structure explicitly identified as missing above. A single closed form for $k(p_1, \ldots, p_n, c)$ analogous to the homogeneous case's $k = 1 + 5p$ is not claimed: assembling one means summing over all $n!$ weighted orderings combined with the now state-dependent timing, which may not collapse as cleanly as the homogeneous case did. What is resolved is the structural piece named above as missing.

C. Terminal commitment

Everywhere above the action set contracts as a side effect of pulling arms. A distinct and practically important variant adds an explicit *terminal action*: after a finite budget of irreversible experiments the learner must declare an operating envelope, and that declaration bounds every subsequent action. This connects to *real options theory* (Dixit and Pindyck, 1994), which values the option to delay an irreversible investment under uncertainty and shows that irreversibility alone creates value in waiting. Our setting differs in the direction of the commitment: the classical real-options question is *when* to make a single irreversible investment; ours is how to allocate a *budget of experiments* whose results jointly determine a permanent, multi-dimensional commitment (the declared envelope), so the relevant decision is an allocation problem nested inside an eventual stopping decision, not a pure timing problem.

The motivating case is design-space filing under ICH Q8. A finite quantity of active ingredient supports a finite number of process runs; each either demonstrates a parameter setting viable or produces an out-of-spec batch. At filing a design space is declared, and thereafter operating inside it requires no regulatory approval while operating outside triggers a variation procedure. The declared envelope is a permanent commitment made on incomplete evidence.

Model. Settings are indexed $0, \ldots, n-1$ with a nominal point at the centre. Yield rises monotonically with process intensity and so does failure probability—the $\mathrm{corr}(v, \kappa) > 0$ structure of Section 10. Each experiment either demonstrates viability or produces an out-of-spec result that blocks that setting permanently. The claimable envelope is the widest contiguous interval of demonstrated settings containing the nominal point. In the operating phase the process drifts; landing inside the envelope collects the yield at the realised setting, landing outside forces reversion to nominal and incurs a variation penalty. Operating value is computed exactly from a discretised drift distribution rather than sampled.

Results (300 seeds per cell):

budget	policy	operating value	envelope width	failed runs
2	greedy (highest yield first)	36.908	**1.00**	0.80
2	edge-first (maximise width)	36.908	**1.00**	0.67
2	**expand outward**	**43.271** (+17.2%)	**2.91**	0.09
4	greedy	41.385	3.44	1.06
4	edge-first	36.908 (−10.8%)	**1.00**	1.14
4	**expand outward**	**44.993** (+8.7%)	**4.66**	0.33
6	greedy	45.058	5.35	1.14
6	edge-first	36.908 (−18.1%)	**1.00**	1.36
6	expand outward	45.090 (+0.1%)	6.22	0.61

Two failure modes appear that have no analogue in the sequential setting.

Greedy buys yield it cannot claim. It spends the budget on the highest-yield settings wherever they lie. But an envelope is an interval containing the nominal point, so a demonstrated setting far from centre is worthless unless everything between is also demonstrated. At budget 2 greedy finishes with envelope width 1.00: it demonstrated the best settings and could claim none of them.

Ambition narrows the envelope it targets. The edge-first policy tests the outermost settings precisely to maximise claimable width. It pays the highest failure probability per experiment, and each failure blocks that setting permanently. Its envelope width is 1.00 at every budget, and its performance degrades monotonically as the budget grows: −10.8% at budget 4, −18.1% at budget 6. Spending more in order to widen the envelope makes it narrower. This is a failure mode practitioners describe directly—an aggressive characterisation campaign ending with a narrower filed design space than a conservative one would have produced.

The budget dependence runs the other way. The advantage of the correct policy *decreases* with budget (+17.2%, +8.7%, +0.1%), the opposite of the sequential setting where the gap grows with pool size. The reason is structural: terminal commitment is a problem about scarcity of evidence at the moment of an irreversible decision. With enough evidence the commitment is no longer made under uncertainty and the difficulty dissolves. It binds hardest exactly when experiments are expensive, which is the regime that motivates it.

Theorem 27 (Boundary-only expansion is optimal)**.** *For the terminal-commitment model above, any policy that tests a setting not adjacent to the current claimable envelope is weakly dominated by one that only ever tests settings adjacent to it.*

Proof. Suppose a policy tests a setting s strictly beyond the current envelope $[lo, hi]$, with an untested gap setting g (adjacent to the envelope) still between them. Since outcomes are independent across settings, s succeeding contributes nothing to the claimable envelope unless the gap eventually fills too. Compare to the policy that tests g first instead, deferring s: if g fails, the envelope is capped there regardless of s's outcome, so testing s at all would have been wasted—deferring or skipping it costs nothing and frees a budget slot for reallocation. If g succeeds, the envelope extends past it and s becomes reachable under the same remaining budget as before. Either way testing the adjacent setting first weakly dominates; induction over the decision sequence gives the general claim. □

Verified independently by exact DP: restricting the action set to boundary-adjacent settings only never loses value relative to the fully unrestricted DP optimum, across 42 instances spanning pool size, budget, centre position, and failure-probability steepness—zero gap in every case, not merely close.

What remains open: which direction. Given boundary-only expansion is optimal in structure, the only remaining decision at each step is which side to test. "Always the cheaper (lower failure-probability) side"—the natural first guess, and the rule this section's earlier numbers used—is *not* optimal: checked against exact DP across 20 instances it mismatches the true optimal choice in 15, losing real value in a concrete case (45.15 achievable vs. 41.69 realised). A resolved instance is informative: with failure probabilities 0.042 (left) and 0.266 (right), the true optimal first move is the *riskier* side, because it also carries higher yield. Scoring each side by the expected value of a successful test, $(1-p_{\text{fail}}) \times$ yield, rather than by failure probability alone, closes most of the gap (9/20 mismatches, and the table above already reflects this improved rule) but not all of it—the true optimal direction appears to depend on remaining budget in a way no static index captures, so a fully general characterisation remains open.

D. Software: the ATTRITION testbed

The accompanying library, ATTRITION (Agent Testbed for Task, Resource and Irreversible-Tradeoff Investigation), is released openly and is the artifact through which every claim in this paper can be checked.

Design. The core is dependency-free apart from NumPy. Gym- and PettingZoo-compatible interfaces are implemented natively rather than by importing either framework, so the package carries no reinforcement-learning dependency while matching both signatures.

Observations withhold the externality. By Theorem 6 the externality coefficients cannot be reliably estimated from experience. Exposing them in the observation vector would therefore let an agent solve a different problem from the one the theory describes, so the default observation contains availability, value estimates and pull counts only. A flag lifts this for oracle experiments.

Scenario packs. Seven named scenarios, each documenting the phenomenon it is intended to exhibit, with the test suite asserting that it does so—a scenario that drifts becomes a detectable regression rather than a silently wrong result.

scenario	std(κ)	corr(v, κ)	greedy loss	greedy regret
high-dispersion	0.6285	+1.000	176.5%	0.00000000
design-space	0.2954	+0.789	99.3%	0.00000000
adaptive-therapy	0.1705	+0.614	36.9%	0.00000000
shared-quota	0.9948	+0.315	3.7%	0.00000000
platform-trial	0.0371	−0.938	**0.0**%	0.00000000
aligned-control	0.6299	−0.971	**0.0**%	0.00000000

This table also makes the alignment condition visible in a way the theory alone does not. The `shared-quota` scenario has the *highest* dispersion in the suite and the *smallest* loss, because its correlation is weakest; the synthetic `aligned-control` scenario has larger dispersion than `adaptive-therapy` and yet no loss at all. Across the suite the loss tracks corr(v, κ) rather than std(κ).

Reproducibility. Every figure regenerates from the library and the exact dynamic-programming reference through a single command. All random seeds are fixed in code. The test suite (107 tests across 60 experiments) contains at least one regression test per theorem and several per result with multiple independently checkable parts, so breaking a theorem's numerical claim breaks a test. Experiments that produced negative or falsified results are retained in the repository rather than removed.

E. Agent tool ecosystems in detail

Figure 6 returns to the routing example of Section 1 and shows the three quantities side by side over a single session. Panel (a) is what an operations dashboard displays; panels (b) and (c) are what is actually happening. The greedy router's regret trace is a flat line at zero for the entire session. Its value per call steps downward three times, and its tool count falls to one. The corrected router shows visible, non-zero regret throughout—and ends the session delivering more value per call with more tools still alive.

This is the practical form of Theorem 1: the monitored quantity carries no signal of the loss, so no alerting threshold defined on it can detect the failure.

F. Correlated-destruction diagnostics: a rollout bug and a scale-fairness artifact

An unreliable Monte Carlo estimate, excluded rather than reported. A Monte Carlo rollout was tested as an independent large-scale reference. It failed a basic sanity check — policy improvement should never make an estimate worse with more compute, and increasing rollout depth moved the estimate substantially in the wrong direction, the signature of an implementation error rather than noise. The cause: planning-phase and actual-trajectory random draws shared one stream, so different compute budgets consumed different amounts of randomness

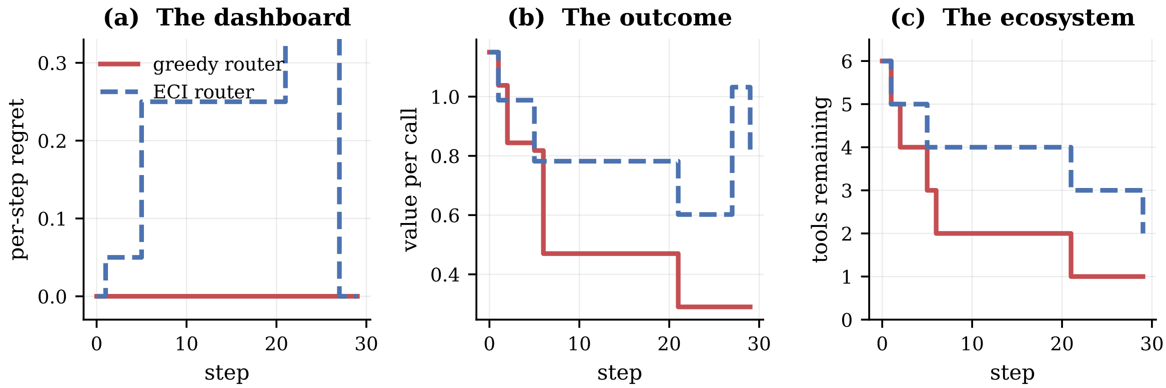

Figure 6: A single routing session, identical seed and tool set. **(a)** The metric: greedy's per-step regret is zero throughout, while the corrected router accrues visible regret. **(b)** Value per call: the router with worse regret delivers more. **(c)** Surviving tools: the router with worse regret preserves more of the ecosystem. The visible regret in (a) is the *price of preservation*, and the standard metric records it as a defect.

before each real step, changing the realised trajectory for reasons unrelated to policy quality. With this corrected, the estimate stabilised but remained theoretically suspect (a valid rollout cannot underperform its base policy), and cross-checking it against exact DP at $n = 6$ showed a 4–5% undershoot even at high depth. The tool is excluded on these grounds; the fix is retained in the released code, but no conclusion is drawn from its output.

Exact computation itself goes further than the original small-scale check used: $n = 12$ runs in roughly 12 seconds. Re-running the full q-sweep there, real ground truth throughout:

q	$V^\star$	greedy	greedy gap	SECI	SECI gap
0.0	11.299	10.482	7.2%	11.096	**1.80%**
0.2	4.171	3.994	4.3%	4.136	**0.84%**
0.4	2.765	2.736	1.1%	2.757	**0.29%**
0.6	1.921	1.916	0.3%	1.919	**0.11%**
0.8	1.512	1.512	0.0%	1.512	**0.03%**

SECI captures 4–5× more of the available opportunity than greedy at every q with meaningful opportunity remaining, confirmed at double the arm count of the original check. The absolute margin between the two shrinks with q because the opportunity itself shrinks — greedy's own gap collapses from 7.2% to 0.0% — not because SECI's relative advantage weakens.

A scale-dependent shortfall, and its cause. At larger scale SECI beats plain ECI everywhere but trails greedy by a small margin in the mid-q range. Varying cluster size and count independently leaves this gap unchanged, ruling out cluster structure as the cause; sweeping n directly shows both policies' values turning *negative* once n exceeds roughly 20 under a δ held fixed across the sweep. The cause is an unfair comparison, not a failure of the correction: total burden capacity $\delta \sum_i e_i$ scales with n while δ was held constant, so the $n = 40$ instances were a strictly harder problem than the $n = 6$ ones. Rescaling δ to hold problem difficulty comparable across scale closes the gap to noise level (-0.029 to $+0.417$ against values of 1.17–37.46, i.e. under 0.2% except where SECI wins outright), while SECI continues to beat plain ECI at every q without exception.

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